

Japan Technology: IT Services: NC company results read-across (5) (manufacturing/automotive): Systema, Argo Graphics, DIT

In this report, we consider the implications for our IT services sector coverage from visits we made to three Not Covered companies (we have no investment views on these companies) in the manufacturing/automotive system integrator space:

Systema (in-car systems/SI), Argo Graphics (automotive CAD/semiconductor-related systems), and Digital Information Technologies (DIT; in-car systems/SI). At **Systema**, the favorable business environment continues, centered on development projects for the mobility and financial sectors, but the company sees FY3/27 operating profits growing only modestly, due to an increase in share-based compensation costs from the issuance of new stock acquisition rights. At **Argo Graphics**, FY3/27 operating profit guidance is for a yoy decline, due to investment curbs by major automotive customers and higher depreciation costs for a data center. At **DIT**, 3Q (Jan-Mar) operating profits declined for the second consecutive quarter, due to investment curbs by multiple automotive customers, and were below guidance.

In terms of read-across, the declining investment appetite in the automotive sector is a slight negative for the overall IT services sector demand environment. While there are no major system integrators with high sales exposure to the automotive sector, we will be closely watching trends at system integrators that serve a wide range of industries, such as Fujitsu (Buy) and NEC (Buy). Below we outline our key takeaways from company commentary (see [Exhibit 1/Exhibit 2/Exhibit 3/Exhibit 4](#) for earnings tables).

Systema (2317.T) (in-car systems/SI)

- (1) **Guidance:** FY3/27 operating profit guidance is for modest growth to ¥16.0 bn (+4% yoy). The main reason for this is an increase in share-based compensation costs (+¥0.84 bn yoy) from the issuance of new stock acquisition rights. Excluding this impact and depreciation, the company forecasts that EBITDA will continue to grow by double digits (%) at +11% yoy. Systema expects the business solutions business to see a decline in sales and profits, due to the peaking out of the PC replacement cycle. However, excluding this business, it forecasts sales and profits to increase, with the favorable business environment for the core development business expected to continue (the company noted the impact from the situation in the Middle East is limited).
- (2) **Next-generation mobility business:** Automakers' investment appetite for

Chikai Tanaka, CFA
+81(3)4587-9840 |
chikai.tanaka@gs.com
Goldman Sachs Japan Co., Ltd.

Yuki Sato
+81(3)4587-8536 | yuki.z.sato@gs.com
Goldman Sachs Japan Co., Ltd.

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Software Defined Vehicles (SDVs) is strong, with demand for ADAS/autonomous driving and digital keys in particular expanding. However, Systema said investment stances vary among customers. Looking at the industry as a whole, the investment appetite of some automakers is waning, which needs to be watched closely.

- (3) **Digital integration business:** Results remain strong, with a high level of inquiries for projects such as core system renewals. By industry, the financial sector is seeing particularly strong growth. Given order trends and other factors, the company said it expects this trend to continue.
- (4) **Medium-term plan:** The operating profit target for the final year, FY3/29, is ¥20.2 bn (implying a +9% CAGR from FY3/26 to FY3/29). While Systema aims to grow all core businesses, it expects the digital integration business to drive growth as it continues to see strong inquiries for core system renewals and other projects. According to the company, this target does not factor in efficiency improvements from data-driven management or reallocation of human resources, which could be an upside factor for earnings.

Argo Graphics (7595.T) (automotive/semiconductors)

- (1) **Guidance:** FY3/27 operating profit guidance is ¥10.3 bn (-4% yoy), which would mark the first yoy decline in six years since the COVID-19 pandemic. Although investment by major semiconductor customers is showing signs of recovery, the company expects investment by major automotive customers to be curbed, and depreciation costs for the data center (DC) that began operations in Kitami City, Hokkaido, in April will increase. The company expects annual depreciation for the DC to be over ¥0.4 bn, with the expense to be incurred from 1Q (Apr-Jun). According to the company, a managed service that provides systems the company owns as fixed assets will start in 1Q, and depreciation for these systems will be incurred upfront.
- (2) **System products business (non-consolidated):** FY3/26 sales were down -0.5% yoy. Within this, PLM sales were up +2% yoy (sluggish growth in automotive), High Performance Computing (HPC) sales were up +12% yoy (contribution from large projects), and IT sales were down -13% yoy (decline for semiconductor sector). For FY3/27, the company forecasts a slight increase in PLM sales, a decrease in HPC sales, and an increase in IT sales. According to the company, (a) in addition to weak earnings across the auto industry, IT investment is being curbed, due to customer-specific factors (strategic changes), making the outlook for the auto business still challenging; (b) on the other hand, multiple major customers are planning to expand investment in semiconductors (at their Hokkaido/Kyushu/Mie plants).

Digital Information Technologies (3916.T) (in-car systems)

- (1) **3Q6/26 (Jan-Mar):** Operating profits declined for the second consecutive quarter, falling by -5% yoy to ¥0.88 bn. This was attributable to the drop-out of large-scale projects and moves to curb investment in the high-margin automotive business. Profits were below guidance, mainly in the embedded solutions and

product solutions businesses. The company left full-year guidance unchanged, implying 4Q operating profit guidance of ¥0.6 bn (+22% yoy) and the need to achieve a substantial increase in profits. DIT said that while achieving sales guidance would be difficult, it aims to achieve profit guidance by curbing strategic investments (development of security products, AI investment, internal system investment, etc.) and bonuses.

- (2) **Embedded solutions business (embedded systems):** 3Q sales and gross profits turned down yoy by -1% and -5%, respectively. In addition to the pullback from large-scale projects booked in 3Q6/25, the investment appetite of major automotive customers is declining rapidly. Both the development and verification businesses are thus facing strong headwinds. According to the company, this trend is likely to continue into FY6/27, and it will need to be offset by the business solutions business (development and maintenance of business systems).

Exhibit 1: Systema (2317.T): Earnings by business

(mn yen)				CoE				
Systema (2317)	24/3	25/3	26/3	27/3E	26/3	Q2	Q3	Q4
	Q1							
Order	46,840	55,412	60,521		15,576	14,616	14,564	15,765
yoy	-	18.3%	9.2%		9.4%	10.9%	12.0%	5.2%
Order Backlog	17,487	20,722	23,785		22,364	22,817	23,022	23,785
yoy	-	18.5%	14.8%		15.7%	15.9%	18.7%	14.8%
Sales	76,940	83,621	94,400	98,000	22,553	24,414	23,096	24,337
yoy	3.2%	8.7%	12.9%	3.8%	14.2%	20.0%	12.8%	5.6%
Gross Profit	18,078	20,978	24,813		5,951	6,254	6,269	6,339
yoy	-1.7%	16.0%	18.3%		21.0%	24.2%	17.9%	11.0%
% of sales	23.5%	25.1%	26.3%		26.4%	25.6%	27.1%	26.0%
SGA	8,364	8,910	9,446		2,435	2,176	2,300	2,535
yoy	-2.2%	6.5%	6.0%		7.1%	3.4%	5.7%	7.6%
% of sales	10.9%	10.7%	10.0%		10.8%	8.9%	10.0%	10.4%
Operating Profits	9,713	12,067	15,367	15,960	3,516	4,077	3,970	3,804
yoy	-1.3%	24.2%	27.3%	3.9%	33.0%	39.1%	26.4%	13.5%
% of sales	12.6%	14.4%	16.3%	16.3%	15.6%	16.7%	17.2%	15.6%
<Sales and OP by Main Segments>								
Next-generation Mobility Business								
Order		7,511	8,600		2,316	1,759	2,441	2,084
yoy		-	14.5%		3.0%	9.5%	43.6%	6.5%
Order Backlog		3,051	4,082		3,581	3,517	4,090	4,082
yoy		-	33.8%		59.9%	41.5%	52.2%	33.8%
Sales		5,540	7,569		1,785	1,821	1,870	2,093
yoy		-	36.6%		64.4%	33.8%	25.0%	31.1%
% of sales		6.6%	8.0%		7.9%	7.5%	8.1%	8.6%
Operating Profits		1,964	3,219		731	780	801	907
yoy		-	63.9%		102.5%	46.6%	54.3%	64.3%
% of sales		35.5%	42.5%		41.0%	42.8%	42.8%	43.3%
Project Management Design Business								
Order		16,628	15,597		4,561	4,324	3,251	3,461
yoy		-	-6.2%		8.0%	3.0%	-14.4%	-21.5%
Order Backlog		5,864	6,163		6,459	6,912	6,401	6,163
yoy		-	5.1%		23.0%	24.4%	16.8%	5.1%
Sales		15,661	15,296		3,964	3,870	3,762	3,700
yoy		-	-2.3%		2.6%	-0.6%	-2.9%	-8.2%
% of sales		18.7%	16.2%		17.6%	15.9%	16.3%	15.2%
Operating Profits		2,583	3,342		848	850	873	771
yoy		-	29.4%		46.7%	41.7%	35.3%	1.4%
% of sales		16.5%	21.8%		21.4%	22.0%	23.2%	20.8%
Digital Integration Business								
Order		9,002	11,145		2,640	2,603	2,735	3,167
yoy		-	23.8%		22.5%	28.1%	22.6%	22.6%
Order Backlog		3,692	4,430		3,948	3,996	4,090	4,430
yoy		-	20.0%		9.4%	14.4%	18.4%	20.0%
Sales		8,815	10,406		2,382	2,556	2,641	2,827
yoy		-	18.0%		16.1%	18.9%	16.5%	20.5%
% of sales		10.5%	11.0%		10.6%	10.5%	11.4%	11.6%
Operating Profits		1,954	2,476		539	666	662	609
yoy		-	26.7%		34.4%	40.5%	18.9%	16.7%
% of sales		22.2%	23.8%		22.6%	26.1%	25.1%	21.5%
IT&DX Service Business								
Order		20,609	22,464		5,432	5,440	5,537	6,055
yoy		-	9.0%		3.8%	10.0%	11.7%	10.7%
Order Backlog		7,257	7,838		7,438	7,454	7,470	7,838
yoy		-	8.0%		0.7%	2.0%	5.8%	8.0%
Sales		20,753	22,356		5,375	5,542	5,631	5,808
yoy		-	7.7%		7.6%	9.1%	6.9%	7.3%
% of sales		24.8%	23.7%		23.8%	22.7%	24.4%	23.9%
Operating Profits		2,848	3,146		678	832	801	835
yoy		-	10.5%		6.6%	24.9%	8.0%	3.9%
% of sales		13.7%	14.1%		12.6%	15.0%	14.2%	14.4%
Business Solution Business								
Order		1,662	2,215		624	491	600	500
yoy		-	33.3%		67.0%	22.4%	84.9%	-11.1%
Order Backlog		861	994		935	935	969	994
yoy		-	15.4%		11.3%	10.2%	35.9%	15.4%
Sales		29,795	35,584		8,453	9,613	8,711	8,807
yoy		-	19.4%		19.4%	41.5%	22.6%	-0.1%
% of sales		35.6%	37.7%		37.5%	39.4%	37.7%	36.2%
Operating Profits		2,274	2,957		707	859	828	563
yoy		-	30.0%		17.2%	75.7%	38.2%	-3.4%
% of sales		7.6%	8.3%		8.4%	8.9%	9.5%	6.4%
Other Businesses & Adjustment								
Sales		3,057	3,189		594	1,012	481	1,102
Operating Profits		444	227		13	90	5	119

Segment change from FY3/26

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 2: Argo Graphics (7595.T): Earnings by business

(mn yen)				CoE	26/3			
ArgoGraphics (7595)	24/3	25/3	26/3	27/3E	Q1	Q2	Q3	Q4
Sales	59,511	69,542	71,526	72,600	16,756	17,024	15,113	22,633
yoy	11.6%	16.9%	2.9%	1.5%	-6.9%	9.1%	-7.8%	15.9%
Gross Profit	16,017	17,868	18,780	19,300	4,367	4,411	3,999	6,003
yoy	13.1%	11.6%	5.1%	2.8%	-5.3%	12.4%	-7.4%	19.6%
% of sales	26.9%	25.7%	26.3%	26.6%	26.1%	25.9%	26.5%	26.5%
SGA	6,843	7,669	8,035	9,000	1,925	1,814	2,071	2,226
yoy	7.1%	12.1%	4.8%	12.0%	9.2%	-1.8%	18.8%	-3.9%
% of sales	11.5%	11.0%	11.2%	12.4%	11.5%	10.7%	13.7%	9.8%
Operating Profit	9,173	10,200	10,745	10,300	2,443	2,597	1,928	3,777
yoy	18.0%	11.2%	5.3%	-4.1%	-14.2%	25.0%	-25.1%	39.8%
% of sales	15.4%	14.7%	15.0%	14.2%	14.6%	15.3%	12.8%	16.7%
<Parent company>								
Sales	48,723	58,251	59,658	60,000	13,641	14,391	12,543	19,083
yoy	11.2%	19.6%	2.4%	0.6%	-10.3%	10.7%	-8.6%	16.9%
Gross Profit	11,555	12,906	14,026	14,400	3,091	3,301	2,931	4,703
yoy	15.8%	11.7%	8.7%	2.7%	-8.8%	18.9%	-4.5%	28.1%
% of sales	23.7%	22.2%	23.5%	24.0%	22.7%	22.9%	23.4%	24.6%
SGA	4,368	4,809	5,378	6,200	1,228	1,170	1,416	1,564
yoy	10.6%	10.1%	11.8%	15.3%	15.4%	4.7%	34.9%	-0.8%
% of sales	9.0%	8.3%	9.0%	10.3%	9.0%	8.1%	11.3%	8.2%
Operating Profit	7,187	8,097	8,648	8,200	1,863	2,131	1,514	3,140
yoy	19.2%	12.7%	6.8%	-5.2%	-19.9%	28.5%	-25.0%	49.9%
% of sales	14.8%	13.9%	14.5%	13.7%	13.7%	14.8%	12.1%	16.5%
I System Product								
Sales	23,433	30,951	30,801		5,122	6,763	6,905	12,011
yoy	12.2%	32.1%	-0.5%		-21.5%	3.4%	-15.4%	23.5%
% of sales	48.1%	53.1%	51.6%		37.5%	47.0%	55.1%	62.9%
(1) PLM Sales	11,150	10,564	10,827		2,523	3,550	2,264	2,490
yoy	49.2%	-5.3%	2.5%		6.2%	65.8%	-31.0%	-10.0%
(2) HPC (High Performance Computing)	3,975	8,968	10,037		1,407	1,418	1,707	5,505
yoy	-26.3%	125.6%	11.9%		-57.6%	-6.6%	72.4%	75.4%
(3) IT sales (Server & Storage, and others)	8,308	11,419	9,936		1,191	1,794	2,936	4,015
yoy	3.7%	37.4%	-13.0%		44.0%	-37.7%	-24.5%	5.0%
Gross Profit	5,264	6,473	6,769		1,213	1,366	1,457	2,733
yoy	27.7%	23.0%	4.6%		-16.6%	13.2%	-14.3%	29.5%
% of sales	22.5%	20.9%	22.0%		23.7%	20.2%	21.1%	22.8%
II Service								
Sales	25,290	27,299	28,857		8,519	7,628	5,637	7,073
yoy	10.2%	7.9%	5.7%		-1.8%	18.0%	1.3%	7.2%
% of sales	51.9%	46.9%	48.4%		62.5%	53.0%	44.9%	37.1%
(1) SW Maintenance	14,537	15,402	16,247		5,505	4,407	3,034	3,301
yoy	14.9%	6.0%	5.5%		3.5%	13.4%	-2.5%	7.0%
(2) HW Maintenance	1,983	2,626	2,820		642	982	471	725
yoy	-5.0%	32.4%	7.4%		-32.4%	57.9%	7.3%	17.9%
(3) System Service	8,125	8,299	8,921		2,191	2,063	1,988	2,679
yoy	8.8%	2.1%	7.5%		1.3%	14.5%	10.0%	6.1%
(4) Development • Education	646	972	869		181	176	144	368
yoy	-12.2%	50.5%	-10.6%		-25.2%	15.0%	-30.1%	-0.8%
Gross Profit	6,291	6,433	7,256		1,878	1,933	1,475	1,970
yoy	7.4%	2.3%	12.8%		-2.9%	23.1%	7.8%	26.2%
% of sales	24.9%	23.6%	25.1%		22.0%	25.3%	26.2%	27.9%
<Subsidiaries >								
Sales	10,788	11,291	11,868	12,600	3,115	2,633	2,570	3,550
yoy	13.4%	4.7%	5.1%	6.2%	11.0%	1.2%	-3.7%	10.6%
Gross Profit	4,462	4,962	4,754	4,900	1,276	1,110	1,068	1,300
yoy	6.7%	11.2%	-4.2%	3.1%	4.6%	-3.3%	-14.4%	-3.4%
% of sales	41.4%	44.0%	40.1%	38.9%	41.0%	42.1%	41.6%	36.6%
SGA	2,475	2,860	2,657	2,800	697	644	655	662
yoy	1.5%	15.5%	-7.1%	5.4%	-0.3%	-11.5%	-5.7%	-10.4%
% of sales	22.9%	25.3%	22.4%	22.2%	22.4%	24.4%	25.5%	18.6%
Operating Profit	1,986	2,103	2,097	2,100	580	466	414	637
yoy	13.9%	5.9%	-0.3%	0.1%	11.1%	10.9%	-25.3%	5.1%
% of sales	18.4%	18.6%	17.7%	16.7%	18.6%	17.7%	16.1%	18.0%

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 3: Digital Information Technologies (DIT) (3916.T): Earnings by business

(mn yen)	CoE			25/6				26/6		
Digital Information Technologies (3916)	24/6	25/6	26/6E	Q1	Q2	Q3	Q4	Q1	Q2	Q3
Sales	19,888	24,159	26,000	5,632	6,175	6,342	6,010	6,246	6,507	6,518
yoy	9.6%	21.5%	7.6%	19.6%	26.9%	22.9%	16.7%	10.9%	5.4%	2.8%
Gross Profit	4,945	6,146		1,349	1,681	1,668	1,449	1,491	1,644	1,697
yoy	14.5%	24.3%		21.4%	38.7%	27.5%	10.2%	10.5%	-2.2%	1.7%
% of sales	24.9%	25.4%		24.0%	27.2%	26.3%	24.1%	23.9%	25.3%	26.0%
SGA	2,520	3,132		708	722	748	954	780	785	820
yoy	10.6%	24.3%		25.3%	29.1%	19.1%	24.2%	10.2%	8.8%	9.6%
% of sales	12.7%	13.0%		12.6%	11.7%	11.8%	15.9%	12.5%	12.1%	12.6%
Operating Profits	2,425	3,014	3,050	641	959	920	494	711	859	876
yoy	18.9%	24.3%	1.2%	17.4%	47.0%	35.2%	-9.5%	10.9%	-10.4%	-4.7%
% of sales	12.2%	12.5%	11.7%	11.4%	15.5%	14.5%	8.2%	11.4%	13.2%	13.4%
<Sales and GP by Segments>										
I Business Solutions Unit										
Sales	11,165	13,198	14,400	3,087	3,368	3,387	3,356	3,421	3,451	3,659
yoy	7.9%	18.2%	9.1%	17.2%	23.2%	18.0%	14.7%	10.8%	2.5%	8.0%
% of sales	56.1%	54.6%	55.4%	54.8%	54.5%	53.4%	55.8%	54.8%	53.0%	56.1%
Business Systems Development	6,243	7,764	8,600	1,804	2,013	2,047	1,900	1,987	2,019	2,203
yoy	9.1%	24.4%	10.8%	25.0%	32.3%	27.9%	13.2%	10.1%	0.3%	7.6%
Operational Support	4,922	5,434	5,800	1,283	1,354	1,340	1,457	1,433	1,432	1,456
yoy	6.4%	10.4%	6.7%	7.9%	11.6%	5.4%	16.7%	11.7%	5.8%	8.7%
Gross Profit	2,852	3,347		764	848	871	864	856	896	979
yoy	24.8%	17.4%		20.1%	19.8%	18.2%	12.1%	12.0%	5.7%	12.4%
% of sales	25.5%	25.4%		24.7%	25.2%	25.7%	25.7%	25.0%	26.0%	26.8%
Business Systems Development	1,646	2,042		453	554	528	507	502	526	591
yoy	38.7%	24.1%		30.9%	33.8%	25.4%	9.0%	10.8%	-5.1%	11.9%
% of sales	26.4%	26.3%		25.1%	27.5%	25.8%	26.7%	25.3%	26.1%	26.8%
Operational Support	1,205	1,305		310	295	343	357	354	369	389
yoy	9.8%	8.3%		6.9%	0.3%	8.5%	17.0%	14.2%	25.1%	13.4%
% of sales	24.5%	24.0%		24.2%	21.8%	25.6%	24.5%	24.7%	25.8%	26.7%
II Embedded Solutions Unit										
Sales	6,897	7,816	8,400	1,861	1,946	2,118	1,891	2,002	2,239	2,091
yoy	10.3%	13.3%	7.5%	13.3%	15.9%	12.8%	11.4%	7.6%	15.1%	-1.3%
% of sales	34.7%	32.4%	32.3%	33.0%	31.5%	33.4%	31.5%	32.1%	34.4%	32.1%
Embedded Product Development	5,098	5,492	6,000	1,363	1,352	1,468	1,309	1,367	1,633	1,543
yoy	11.0%	7.7%	9.2%	12.8%	9.9%	5.4%	3.3%	0.3%	20.8%	5.1%
Embedded Product Verification	1,799	2,324	2,400	497	595	650	582	635	606	548
yoy	8.3%	29.2%	3.3%	14.5%	32.2%	34.6%	34.7%	27.8%	1.8%	-15.7%
Gross Profit	2,228	2,507		587	629	692	599	609	681	656
yoy	9.8%	12.5%		10.1%	16.3%	13.3%	10.3%	3.7%	8.3%	-5.2%
% of sales	32.3%	32.1%		31.5%	32.3%	32.7%	31.7%	30.4%	30.4%	31.4%
Embedded Product Development	1,690	1,795		443	445	485	422	414	500	501
yoy	10.6%	6.2%		9.9%	9.6%	5.2%	0.5%	-6.5%	12.4%	3.3%
% of sales	33.2%	32.7%		32.5%	32.9%	33.0%	32.2%	30.3%	30.6%	32.5%
Embedded Product Verification	537	711		144	183	207	177	194	181	156
yoy	7.0%	32.4%		10.8%	36.6%	38.0%	43.9%	34.7%	-1.1%	-24.6%
% of sales	29.8%	30.6%		29.0%	30.8%	31.8%	30.4%	30.6%	29.9%	28.5%
III Product Solutions Unit										
Sales	1,095	2,277	2,300	493	664	605	515	566	573	528
yoy	27.9%	107.9%	1.0%	121.1%	136.3%	155.3%	45.5%	14.8%	-13.7%	-12.7%
% of sales	5.5%	9.4%	8.8%	8.8%	10.8%	9.5%	8.6%	9.1%	8.8%	8.1%
Gross Profit	394	917		207	319	220	171	220	226	196
yoy	20.1%	132.7%		102.9%	243.0%	152.9%	52.7%	6.3%	-29.2%	-10.9%
% of sales	36.0%	40.3%		42.0%	48.0%	36.4%	33.2%	38.9%	39.4%	37.1%
IV System sales										
Sales	729	866	900	189	198	231	247	256	244	240
yoy	5.8%	18.9%	3.9%	-9.2%	15.2%	31.3%	43.8%	35.1%	22.9%	3.8%
% of sales	3.7%	3.6%	3.5%	3.4%	3.2%	3.6%	4.1%	4.1%	3.7%	3.7%
Gross Profit	268	285		65	68	82	52	86	91	88
yoy	4.7%	6.3%		-18.8%	9.7%	20.6%	-10.3%	32.3%	33.8%	7.3%
% of sales	36.8%	32.9%		34.3%	34.3%	35.5%	21.0%	33.6%	37.3%	36.7%

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 4: IT services sector earnings outlook

(mn yen)													
Company	Results (FY ended March)	25/3	26/3	GSE			CoE 27/3	CY25				CY26	
				27/3	28/3	29/3		1-3	4-6	7-9	10-12	1-3	4-6E
【Mid-sized System Integrators】													
2327 NS Solutions (NSSOL)	Sales	338,301	381,340	416,000	431,000	453,000	417,000	98,007	82,684	95,691	97,020	105,945	95,000
	%y-y	8.9%	12.7%	9.1%	3.6%	5.1%	9.4%	8.3%	7.6%	19.9%	15.9%	8.1%	14.9%
	OP	38,497	44,242	47,500	50,000	54,000	47,500	8,910	8,485	9,787	12,718	13,252	8,500
	%y-y	10.0%	14.9%	7.4%	5.3%	8.0%	7.4%	-23.9%	-3.7%	5.1%	11.0%	48.7%	0.2%
3626 TIS	OP margin	11.4%	11.6%	11.4%	11.6%	11.9%	11.4%	9.1%	10.3%	10.2%	13.1%	12.5%	8.9%
	Sales	571,687	596,479	622,500	642,500	658,500	620,000	155,126	140,316	148,209	147,726	160,228	146,830
	%y-y	4.1%	4.3%	4.4%	3.2%	2.5%	3.9%	7.5%	4.7%	4.7%	4.8%	3.3%	4.6%
	OP	69,047	76,229	81,500	86,000	89,300	81,000	20,163	16,353	19,200	19,250	21,426	17,200
4307 Nomura Research Institute (NRI)	%y-y	6.9%	10.4%	6.9%	5.5%	3.8%	6.3%	18.7%	16.3%	16.7%	4.8%	6.3%	5.2%
	OP margin	12.1%	12.8%	13.1%	13.4%	13.6%	13.1%	13.0%	11.7%	13.0%	13.0%	13.4%	11.7%
	Sales	764,813	814,708	854,200	897,200	941,400	850,000	196,567	195,770	201,295	205,268	212,374	205,900
	%y-y	3.8%	6.5%	4.8%	5.0%	4.9%	4.3%	5.4%	4.1%	6.7%	7.2%	8.0%	5.2%
4684 Obic	OP	134,907	58,273	178,000	198,000	219,000	175,000	32,554	37,246	41,553	39,981	-60,506	41,000
	%y-y	12.0%	-56.8%	205.5%	11.2%	10.6%	200.3%	11.6%	14.1%	26.0%	8.9%	N.M.	10.1%
	OP margin	17.6%	7.2%	20.8%	22.1%	23.3%	20.6%	16.6%	19.0%	20.6%	19.5%	-28.5%	19.9%
	Sales	121,240	135,209	149,000	161,800	174,000	148,700	31,517	32,431	33,353	34,329	35,096	35,860
4733 Obic Business Consultants (OBC)	%y-y	8.6%	11.5%	10.2%	8.6%	7.5%	10.0%	12.5%	12.7%	9.7%	12.4%	11.4%	10.6%
	OP	78,378	88,823	99,300	108,800	118,000	98,000	19,812	21,479	22,176	22,610	22,558	24,100
	%y-y	10.5%	13.3%	11.8%	9.6%	8.5%	10.3%	13.8%	15.2%	11.0%	13.4%	13.9%	12.2%
	OP margin	64.6%	65.7%	66.6%	67.2%	67.8%	65.9%	62.9%	66.2%	66.5%	65.9%	64.3%	67.2%
8056 BIPROGY (ex Nihon Unisys)	Sales	46,984	51,401	57,650	64,300	71,670	57,500	12,220	12,151	12,781	12,946	13,524	13,370
	%y-y	12.0%	9.4%	12.2%	11.5%	11.5%	11.9%	6.1%	8.2%	11.0%	7.7%	10.7%	10.0%
	OP	21,744	23,581	26,800	31,000	35,800	26,500	5,833	5,570	5,590	6,110	6,311	6,200
	%y-y	16.0%	8.4%	13.7%	15.7%	15.5%	12.4%	6.3%	7.5%	13.4%	5.3%	8.2%	11.3%
[Large System Integrators]	OP margin	46.3%	45.9%	46.5%	48.2%	50.0%	46.1%	47.7%	45.8%	43.7%	47.2%	46.7%	46.4%
	Sales	404,010	433,686	484,500	516,200	546,900	470,000	124,745	96,844	108,288	101,712	126,842	107,100
	%y-y	9.2%	7.3%	11.7%	6.5%	5.9%	8.4%	12.1%	9.6%	10.9%	9.1%	1.7%	10.6%
	OP	39,066	42,604	48,400	54,600	60,800	48,400	13,900	8,569	12,781	8,998	12,256	8,800
6501 Hitachi Digital Systems & Services (DSS)	%y-y	17.4%	9.1%	13.6%	12.8%	11.4%	13.6%	51.9%	30.3%	12.4%	24.6%	-11.8%	2.7%
	OP margin	9.7%	9.8%	10.0%	10.6%	11.1%	10.3%	11.1%	8.8%	11.8%	8.8%	9.7%	8.2%
	Sales	2,832,584	2,940,000	3,291,900	3,579,000	3,890,100	3,190,000	823,300	617,400	708,800	714,900	898,912	668,650
	%y-y	9.0%	3.8%	12.0%	8.7%	8.7%	8.5%	6.2%	-2.6%	4.5%	2.6%	9.2%	8.3%
6701 NEC	Adjusted OP	374,700	430,000	477,100	524,700	572,900	480,000	124,400	61,800	105,200	116,875	159,200	97,011
	%y-y	20.6%	14.8%	11.0%	10.0%	9.2%	11.6%	19.2%	-16.1%	22.9%	28.6%	28.0%	57.0%
	OP margin	13.2%	14.6%	14.5%	14.7%	14.7%	15.0%	15.1%	10.0%	14.8%	16.3%	17.7%	14.5%
	Sales	3,423,431	3,582,733	3,596,000	3,755,000	3,916,000	3,500,000	1,101,622	715,658	854,121	858,000	1,160,416	740,000
6702 Fujitsu	%y-y	-1.5%	4.7%	0.4%	4.4%	4.3%	-2.3%	1.6%	3.7%	7.3%	2.7%	5.3%	3.4%
	OP	256,497	359,913	419,000	467,000	518,000		130,331	35,389	83,181	66,591	174,752	39,000
	%y-y	36.4%	40.3%	16.4%	11.5%	10.9%		10.3%	679.5%	107.2%	-18.3%	34.1%	10.2%
	OP margin	7.5%	10.0%	11.7%	12.4%	13.2%		11.8%	4.9%	9.7%	7.8%	15.1%	5.3%
6702 Fujitsu	Non-GAAP OP	311,301	397,234	443,000	490,000	540,000	420,000	148,966	39,995	87,494	82,382	187,363	45,000
	%y-y	36.8%	27.6%	11.5%	10.6%	10.2%	5.7%	16.2%	145.4%	80.2%	-15.5%	25.8%	12.5%
	OP margin	9.1%	11.1%	12.3%	13.0%	13.8%	12.0%	13.5%	5.6%	10.2%	9.6%	16.1%	6.1%
	Sales	3,550,116	3,502,971	3,484,000	3,667,000	3,849,000	3,510,000	1,142,134	749,859	816,683	884,642	1,051,787	738,000
6702 Fujitsu	%y-y	2.1%	-1.3%	-0.5%	5.3%	5.0%	0.2%	9.6%	-1.2%	3.0%	3.4%	-7.9%	-1.6%
	OP	265,089	348,329	415,000	483,000	552,000	415,000	159,200	33,486	71,859	105,656	137,328	36,500
	%y-y	77.5%	31.4%	19.1%	16.4%	14.3%	19.1%	37.8%	133.7%	150.7%	68.2%	-13.7%	9.0%
	OP margin	7.5%	9.9%	11.9%	13.2%	14.3%	11.8%	13.9%	4.5%	8.8%	11.9%	13.1%	4.9%
NTT Data (Global Solution segment in NTT)	Adjusted OP	307,265	390,589	425,000	493,000	562,000	425,000	170,184	35,119	86,208	107,836	161,426	39,000
	%y-y	15.8%	27.1%	8.8%	16.0%	14.0%	8.8%	6.9%	111.9%	74.1%	51.9%	-5.1%	11.1%
	OP margin	8.7%	11.2%	12.2%	13.4%	14.6%	12.1%	14.9%	4.7%	10.6%	12.2%	15.3%	5.3%
	Sales	4,638,721	5,004,614	5,206,000	5,451,000	5,682,000	5,190,000	1,230,965	1,104,369	1,256,098	1,283,336	1,360,813	1,166,000
NTT Data (Global Solution segment in NTT)	%y-y	6.2%	7.9%	4.0%	4.7%	4.2%	3.7%	3.3%	-0.7%	11.4%	9.9%	10.5%	5.6%
	OP	323,862	488,211	470,000	530,000	590,000	470,000	87,881	57,786	211,192	115,225	104,008	60,000
	%y-y	4.6%	50.7%	-3.7%	12.8%	11.3%	-3.7%	-21.9%	-1.4%	133.7%	32.5%	18.4%	3.8%
	OP margin	7.0%	9.8%	9.0%	9.7%	10.4%	9.1%	7.1%	5.2%	16.8%	9.0%	7.6%	5.1%
4704 Trend Micro	Gain on sales of DC		129,500	70,000	70,000	70,000	70,000				129,500		
	OP (excl. gain on sales of DC)		358,711	400,000	460,000	520,000	400,000				81,692		
	%y-y		10.8%	11.5%	15.0%	13.0%	11.5%				-9.6%		
	OP margin		7.0%	7.6%	8.3%	9.0%	7.5%				6.5%		
Company	Results (FY ended Dec)	24/12	25/12	GSE		CoE 26/12	CY25				CY26		
				26/12	28/12		1-3	4-6	7-9	10-12	1-3	4-6E	
4704 Trend Micro	Sales	272,638	275,984	300,000	312,500	325,600	301,500	67,501	66,408	68,844	73,231	73,856	72,890
	%y-y	9.6%	1.2%	8.7%	4.2%	4.2%	9.2%	2.4%	-3.2%	1.1%	4.6%	9.4%	9.8%
	OP	48,105	57,777	55,000	58,000	61,200	56,400	15,006	13,466	16,002	13,303	15,558	12,890
	%y-y	47.6%	20.1%	-4.8%	5.5%	5.5%	-2.4%	23.7%	9.4%	8.1%	50.0%	3.7%	-4.3%
4768 Otsuka	OP margin	17.6%	20.9%	18.3%	18.6%	18.8%	18.7%	22.2%	20.3%	23.2%	18.2%	21.1%	17.7%
	Sales	1,107,668	1,322,791	1,348,800	1,391,200	1,451,600	1,311,000	315,541	379,591	310,587	317,072	344,753	376,400
	%y-y	13.3%	19.4%	2.0%	3.1%	4.3%	-0.9%	18.3%	25.3%	23.0%	11.1%	9.3%	-0.8%
	OP	74,360	89,943	88,000	93,000	99,000	90,000	21,175	27,998	17,356	23,414	23,505	27,000
4478 freee	%y-y	18.1%	21.0%	-2.2%	5.7%	6.5%	0.1%	22.8%	29.6%	30.2%	5.5%	11.0%	-3.6%
	OP margin	6.7%	6.8%	6.5%	6.7%	6.8%	6.9%	6.7%	7.4%	5.6%	7.4%	6.8%	7.2%
	Company	Results (FY ended Jun)	24/6	25/6	GSE		CoE 26/6	CY25				CY26	
					26/6	28/6		1-3	4-6	7-9	10-12	1-3	4-6E
4478 freee	Sales	25,431	33,271	42,260	50,880	60,100	41,930	8,600	9,420	9,744	10,198	10,902	11,417
	%y-y	32.3%	30.8%	27.0%	20.4%	18.1%	26.0%	29.3%	34.4%	32.1%	29.5%	26.8%	21.2%
	OP	-8,387	611	1,040	3,300	6,000		316	-573	272	260	91	417
	%y-y	N.M.	N.M.	70.2%	217.3%	81.8%		N.M.	N.M.	27.2%	-60.3%	-71.1%	N.M.
4478 freee	OP margin	-33.0%	1.8%	2.5%	6.5%	10.0%		3.7%	-6.1%	2.8%	2.5%	0.8%	3.7%
	Adjusted OP	-7,562	1,885	2,540	5,100	8,100	2,520	659	-230	690	662	445	743
	%y-y	N.M.	N.M.	34.7%	100.8%	58.8%	33.7%	N.M.	N.M.	43.8%	-32.2%	-32.5%	N.M.
	OP margin	-29.7%	5.7%	6.0%	10.0%	13.5%	6.0%	7.7%	-2.4%	7.1%	6.5%	4.1%	6.5%

NEC's non-GAAP operating profits are operating profits plus amortization of intangible fixed assets of acquired companies and one-off gains/expenses such as restructuring costs. Fujitsu's adjusted operating profits are operating profits plus one-off gains/expenses such as restructuring costs. For reference, we have included adjusted operating profits for Hitachi's Digital System & Service business (equivalent to IT services) (covered by Ryo Harada,

Disclosure Appendix

Reg AC

We, Chikai Tanaka, CFA and Yuki Sato, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Chikai Tanaka, CFA Goldman Sachs Japan Co., Ltd., Yuki Sato Goldman Sachs Japan Co., Ltd..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

Fujitsu (Buy, ¥3,485) and NEC (Buy, ¥4,224)

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

There are no company-specific disclosures for: Argo Graphics Inc. (¥1,247), Digital Information Technology Corporation (¥826) and Systema Corp. (¥416)

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	60%	45%

As of April 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,074 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a

principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Distribution of ratings: See the distribution of ratings disclosure above. **Price chart:** See the price chart, with changes of ratings and price targets in prior periods, above, or, if electronic format or if with respect to multiple companies which are the subject of this report, on the Goldman Sachs website at <https://www.gs.com/research/hedge.html>.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell

on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors.

Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.